

With so many chips, how to manage the e-waste?

If there is going to be a chip in every object, what will happen to the e-waste generated during the manufacture of these chips, and when these objects (and the chips in them) reach end of their life?

"Of course, there have been concerns. But when you consider that many of the IC chips are essentially silicon chemicals, you will realise ordinary stones and sand have silicon composition too," quips Ishikawa.

So the issue is more about the materials used for packaging the ICs. While ceramic-based packaging is quite clean, it is the resin mould based packaging that is a bit suspicious in this regard and needs to be tracked.

"But, in the case of very tiny chips used for IoT applications without much packaging (they are often sealed into sheets or some type of planar structure), the issue may not be that serious," Ishikawa says. He also points out that the tiny chips attached to the objects might actually be beneficial with regards to e-waste—as the chip will help trace the larger object in which it is embedded and organise its proper disposal!

Tushar Abraham, a former CDAC-ian, ubiquitous computing enthusiast and specialist in ZigBee wireless communication, says, "The number of devices in the mobile, personal computing and appliances space is anyway seeing an explosion. Ubi-comp will only integrate these devices in a more meaningful manner. Either way, India is not currently prepared to deal with the huge amount of e-waste that is generated. An effective e-waste policy will have to be put in place to deal with the current and future usage patterns of its 1.2 billion population. I believe the Pollution Control Board is working on such a policy. However, we still do not have companies that have the technology to retrieve valuable metals and other materials from this e-waste. Therefore it is imperative that stringent penalties are imposed on polluters. Tax incentives and FDI must be allowed to bring in cutting-edge technology into the country to meet the ever-growing demand."

with issues of synchronisation and communication. Significant attention will be needed on coordinated multiprocessing—not only between tightly-coupled processors but also multi-computing for loosely-coupled entities. Common languages will have to be developed for cooperative computing, data sharing, coherence and synchronisation.

The capacity requirements of a network are inherently dependent on the type of traffic carried through the various segments from access to backbone. Ubiquitous computing paradigm will be completely different from the current pattern of the traffic based on the use of PC or wireless smartphone like environment. The current model envisages primarily video traffic using packet transport. The estimated bandwidth for a user in the near future is 20 Mbps. The corresponding traffic for ubiquitous computing will be primarily machine-to-machine communications with a large number of small bursts, and it will be an order of magnitude higher. Also, in current networks, the present-day traffic paths are primarily point-to-point or point-to-multipoint. In ubiquitous computing, a fully connected node-to-node network will be needed with broadband

traffic being carried simultaneously between different paths.

The access network of today is primarily based on a node communicating with a set of user devices in multi-access mode. Ubi-comp architecture will have the devices themselves acting as interconnected nodes with mesh networking. Mesh networking entails new domains and challenges in areas of self-configurability and self-healing, and needs to be highly resistant to disruption. Current networks do not generally possess such attributes. Capability to automatically discover and recognise each other is another key attribute required for ubiquitous computing. Also, the failure mode needs to follow a graceful degradation process and default to a safe mode. The network will require universal broadband wireless coverage with no gaps or dead zones.

The usable wireless spectrum will be a major limitation to meet the expected bandwidth requirements. In particular, proximity area connectivity will demand extremely high bandwidth. Some level of mitigation can be done by using cell splitting techniques to smaller and smaller geographical domains using even zepto or yocto

cells instead of pico or femto cells. Additional proximity area capacity may require exploring optical frequencies using networks like Li-Fi.

Unique IDs for all objects in the IoT

If there are going to be billions of objects connected to the IoT, we also need some means to identify all of them uniquely. Considering the 128-bit naming facility provided by IPv6, it appears as if there will be no problem in doing so. However, Ishikawa points out that "IPv6 is only for IP networks." Ubiquitous computing, on the other hand, is going to involve heterogeneous networks, so the unique ID issue still looms large.

"There will be IP-connected portions and non IP-connected portions, which mainly consist of very small chips with very low CPU power. And actually, the number of objects in such a non-IP connected domain may occupy the majority of the whole domain space," explains Ishikawa.

He adds, "For example, food traceability is considered a great use-case for ubi-comp, but we do not expect every single milk carton to carry an IPv6 address. IPv6 is useful when the object is a network node, but milk cartons themselves don't communicate. The reader device for QR codes, say, on milk cartons will speak IPv6. But we certainly need to assign unique numbers to each milk carton for food traceability. At ITU-T, we are investigating what to do with such requirements, and have proposed a standard called H.LDscheme (which will be renamed to H.6421 once it becomes official) to be used for such ubi-comp environments, and made sure such coding can coexist peacefully with legacy identifier systems. Already, a similar coding scheme is used by our team in Japan and partners in Finland."

Calm and easy computing

Does the thought of so many computers and networks make you feel dizzy? Hopefully, ubi-comp would not be that way. A lot of research is underway across the world to ensure that.

According to the YRP-UNL, what